AARUPADAI VEEDU INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SEMESTER END EXAMINATION - 2024-2025

Degree & Branch:	OUTCOMES AND PROGRAMME SPECIFIC OUTCOMES	**Regulations:**	2021 (Autonomous)
Course Code & Title:	35021P28 - R Programming	**Semester / Year:**	IV Semester / 2024-2025
Date & Session:	FN Session	**Duration & Marks:**	90 Minutes | Max. 50 M

Instructions: Answer ALL questions in Part A. Answer either (a) or (b) from each question in Part B. Show calculations clearly.

PART - A

Q.No	Question Text	CO	Bloom's	Marks
1	Part I (Short Answer - 1 Marks): Formulate and address the short answer requirements for Introduction to R in R, detailing the core technical mechanisms. Part II (Mcq - 1 Marks): Extend your solution through mcq, evaluating the working mechanism, performance implications, and practical limitations in R.	CO1	K1 - Remember	2
2	Part I (Short Answer - 1 Marks): Formulate and address the short answer requirements for R Data Structures in R, detailing the core technical mechanisms. Part II (Mcq - 1 Marks): Extend your solution through mcq, evaluating the working mechanism, performance implications, and practical limitations in R.	CO1	K2 - Understand	2
3	Part I (Short Answer - 1 Marks): Formulate and address the short answer requirements for applying functions to matrix rows and columns in R, detailing the core technical mechanisms. Part II (Mcq - 1 Marks): Extend your solution through mcq, evaluating the working mechanism, performance implications, and practical limitations in R.	CO2	K1 - Remember	2
4	Part I (Short Answer - 1 Marks): Formulate and address the short answer requirements for Adding and deleting rows and columns in R, detailing the core technical mechanisms. Part II (Mcq - 1 Marks): Extend your solution through mcq, evaluating the working mechanism, performance implications, and practical limitations in R.	CO2	K2 - Understand	2
5	Part I (Short Answer - 1 Marks): Formulate and address the short answer requirements for Factors and Tables in R, detailing the core technical mechanisms. Part II (Mcq - 1 Marks): Extend your solution through mcq, evaluating the working mechanism, performance implications, and practical limitations in R.	CO3	K3 - Apply	2

PART - B

Q.No	Question Text	CO	Bloom's	Marks
6	Explain the features of R and describe the steps involved in creating, executing and saving an R script with suitable examples.	CO1	K2 - Understand	5
7	Explain the characteristics of lists in R and discuss how they differ from atomic vectors and matrices. Describe the operations used to create, index, and apply functions to recursive lists with code examples.	CO2	K4 - Analyze	5
8	Explain the characteristics of lists in R and discuss how they differ from atomic vectors and matrices. Describe the operations used to create, index, and apply functions to recursive lists with code examples.	CO2	K4 - Analyze	5
9	Explain the internal representation and utility of factors in R. Describe how frequency tables and cross-tabulations are constructed and analyzed using `table()`, `prop.table()`, and `margin.table()` with practical code demonstrations.	CO3	K4 - Analyze	5

PART - C

Q.No	Question Text	CO	Bloom's	Marks
10	Explain the characteristics of lists in R and discuss how they differ from atomic vectors and matrices. Describe the operations used to create, index, and apply functions to recursive lists with code examples.	CO2	K4 - Analyze	10

11	Explain the internal representation and utility of factors in R. Describe how frequency tables and cross-tabulations are constructed and analyzed using `table()`, `prop.table()`, and `margin.table()` with practical code demonstrations.	CO3	K4 - Analyze	10

Course Outcome (CO) & Cognitive Level Distribution Summary

CO	Total Marks	Weightage (%)
CO1	9 M	18.0%
CO2	24 M	48.0%
CO3	17 M	34.0%